Information Engineering“IE —INT 461

2 –Information Strategy planning and data and process modeling

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Outline

1. Chapter Preview and objectives
2. Strategic Planning
3. OVERVIEW OF DATA AND PROCESS MODELING TOOLS
4. Guidelines for Drawing DFDs

The objective of this chapter

- Describe data and process modeling concepts and tools, including data flow diagrams, a data dictionary, and process descriptions.
- Describe work breakdown structures, task patterns, and critical path analysis
- Describe the symbols used in data flow diagrams and explain the rules for their use.
- Draw data flow diagrams in a sequence, from general to specific
- Explain how to level and balance a set of data flow diagrams
- Describe how a data dictionary is used and what it contains
- Use process description tools, including structured English, decision tables, and decision trees
- Describe the relationship between logical and physical models

STRATEGIC PLANNING — A FRAMEWORK FOR IT SYSTEMS DEVELOPMENT

Why does a systems analyst or information engineering need to know about strategic planning?

The answer might be found in an old story about two stone-cutters who were hard at work when a passerby asked them what they were doing. “I am cutting stones,” said the first worker. The second worker replied, “I am building a cathedral.”
So it is with information technology: One analyst might say, “I am using a CASE tool,” whereas another might say, “I am helping the company succeed in a major new business venture.” Systems analysts should focus on the larger, strategic role of IT as they carry out their day-to-day responsibilities.

Strategic Planning definition

Strategic planning

Strategic planning is a process in which an organization's leaders define their **vision** for the future and identify their organization's **goals and objectives**. The process includes establishing the sequence in which those goals should be realized so that the organization can reach its stated vision.
Strategic planning starts with a management review called a SWOT analysis.

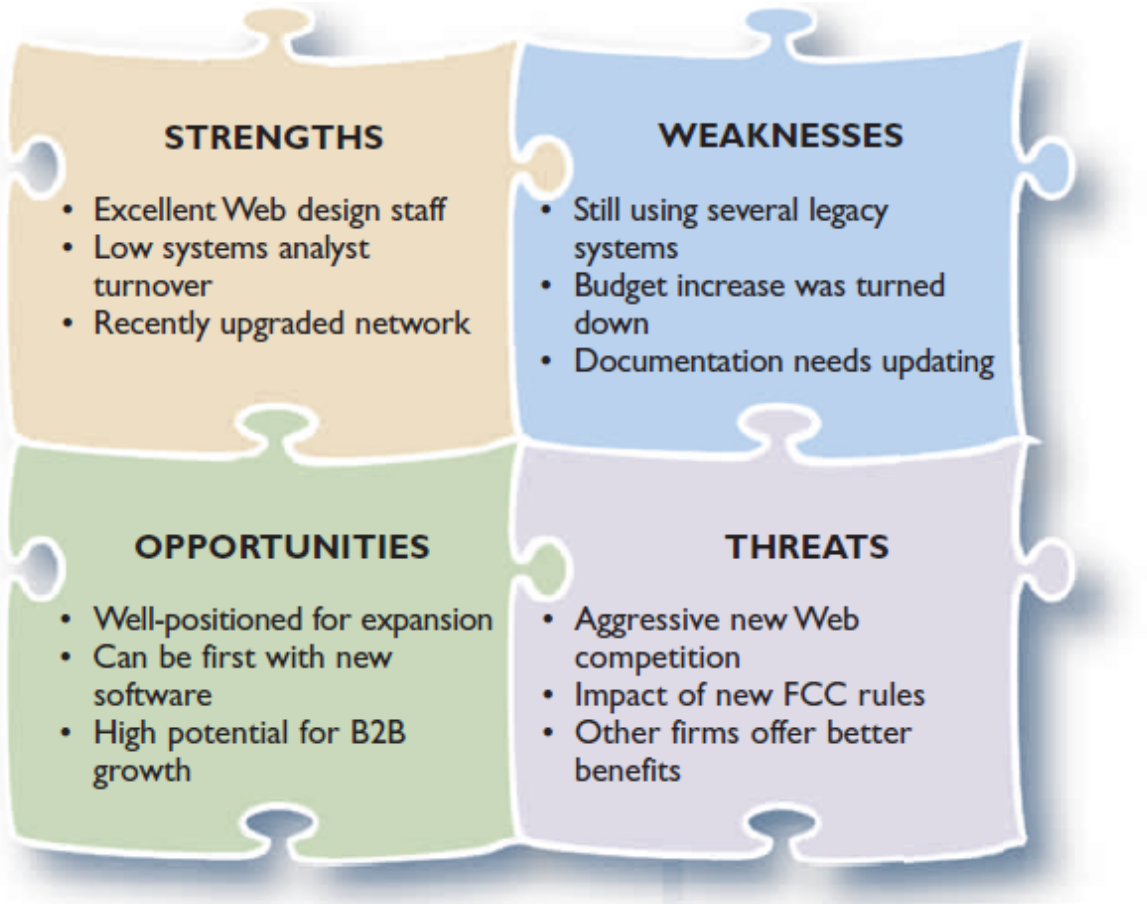
SWOT analysis

The letters **SWOT** stand for strengths, weaknesses, opportunities, and threats. A SWOT analysis usually starts with a broad overview. The first step is for top management to respond to questions like these:

- What are our **S**trengths, and how can we use them to achieve our business goals?
- What are our **W**eaknesses, and how can we reduce or eliminate them?
- What are our **O**pportunities, and how do we plan to take advantage of them?
- What are our **T**hreats, and how can we assess, manage, and respond to the possible risks?

SWOT

A SWOT analysis is a solid foundation for the strategic planning process, because it examines a firm's technical, human, and financial resources. In Figure 2-2, the bulleted lists show samples of typical strengths, weaknesses, opportunities, and threats.



SWOT ANALYSIS

SWOT ANALYSIS



Strength	Our patent covers valuable technology that we can use in many popular products.
Weakness	Our patent, like every patent, has a limited life, and when it expires, the technology will no longer be protected.
Opportunity	We might be able to use the patented technology in more products, license it to other firms, or expand our research to develop new patents based on the original technology.
Threat	There is a chance that a competitor will develop technology that is similar to ours, but does not technically infringe upon our patent.

FIGURE 2-3 Sample SWOT analysis of a specific asset, such as a patent.

Figure: Caption

Frame Title

There is no standard approach to strategic planning. Some managers believe that a firm's mission statement should contain an inspirational message to its stakeholders. Others feel that unless a firm starts with a realistic SWOT assessment, it might develop a mission statement that is unachievable. The majority of companies view the strategic planning process as a dynamic interaction, similar to the diagram in Figure 2-4, where the company's mission statement reflects a long-term horizon, but sets forth goals that are achievable and consistent with real-world conditions. Figure 2-5 on the next page shows how purpose, vision, and values are revealed in one company's Web site.

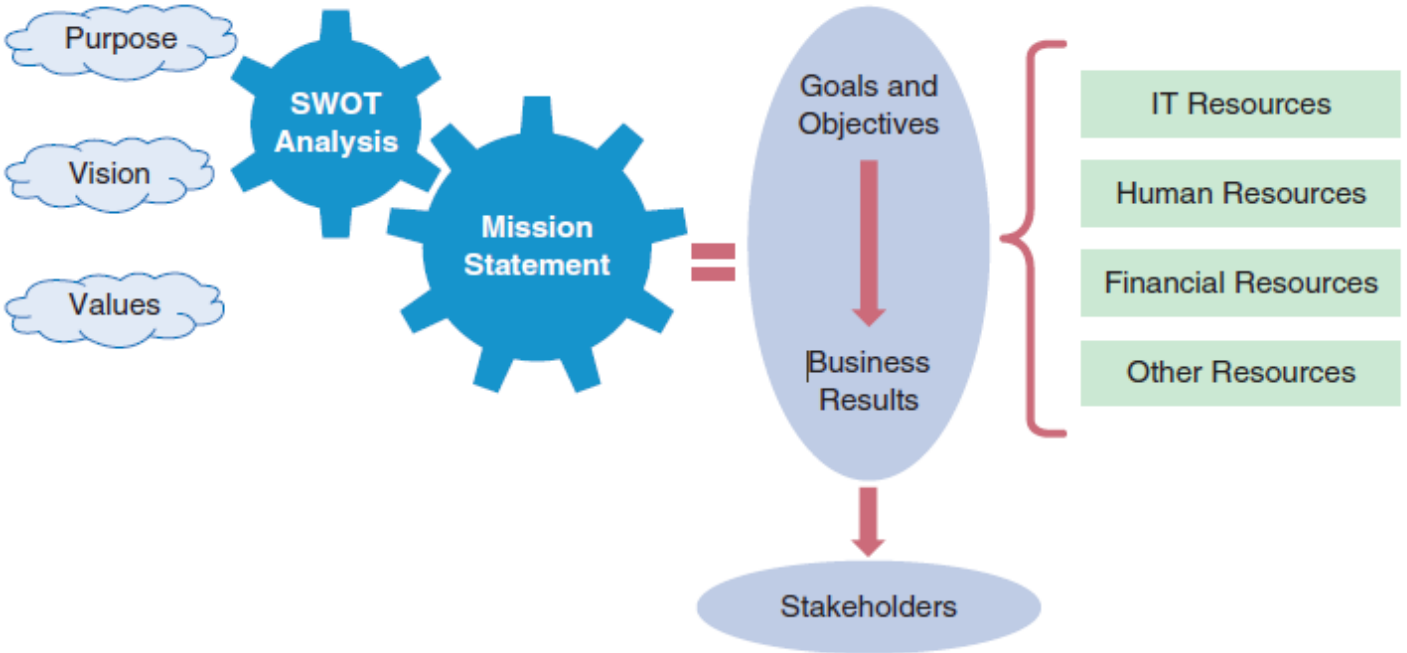
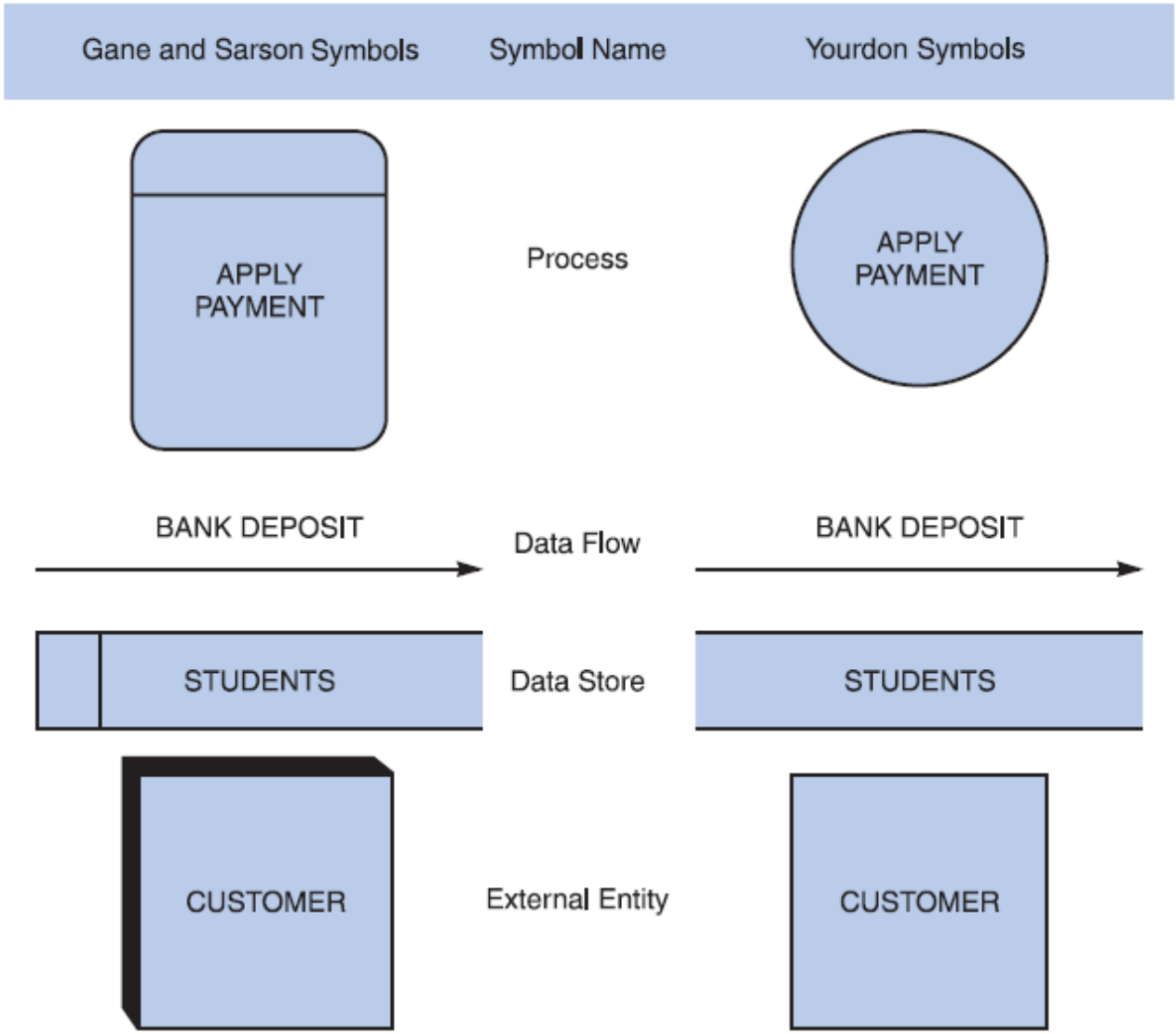


FIGURE 2-4 Strategic planning is a dynamic process that identifies specific goals and objectives that

Data Fellow Diagram

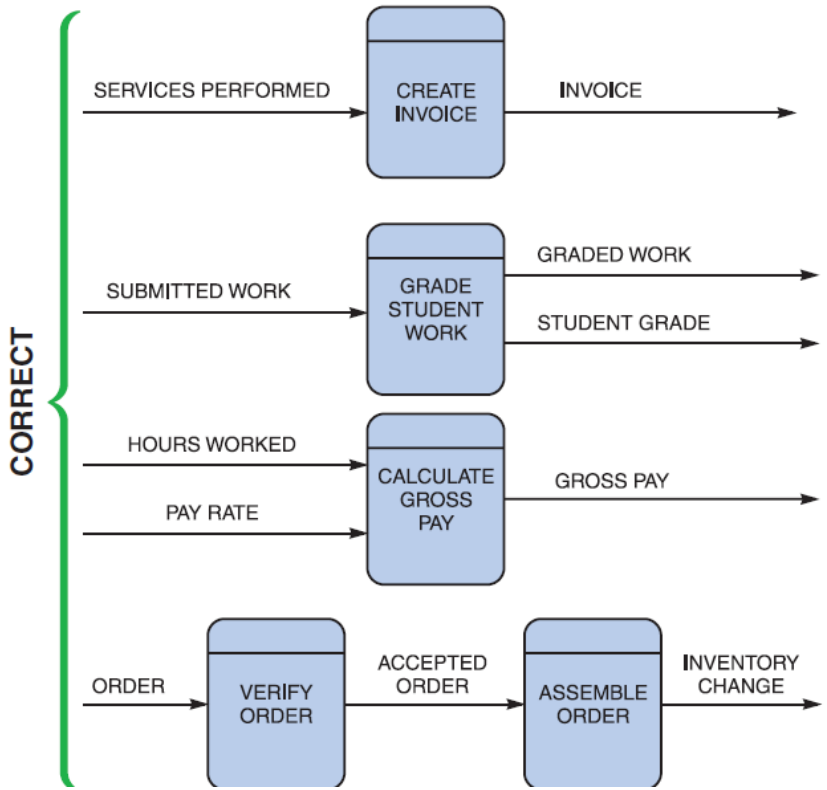
- Systems analysts use many graphical techniques to describe an information system. One popular method is to draw a set of data flow diagrams. A data flow diagram (DFD) uses various symbols to show how the system transforms input data into useful information. Other graphical tools include **object models** and **entity-relationship diagrams**, which are described in Chapter 9 (Data Design).
- A data flow diagram (DFD) shows how data moves through an information system but does not show program logic or processing steps. A set of DFDs provides a logical model that shows what the system does, not how it does it. That distinction is important because focusing on implementation issues at this point would restrict your search for the most effective system design
- DFD Symbols use four basic symbols that represent **processes**, **data flows**, **data stores**, and **entities**. Several different versions of DFD symbols exist, but they all serve the same purpose. DFD examples in this textbook use the Gane and Sarson symbol set. A Symbols are referenced by using all capital letters for the symbol name.

DFDs as a black box



DATA FLOW SYMBOL

- A data flow is a path for data to move from one part of the information system to another.
- A data flow in a DFD represents one or more data items
- A data flow name consists of a singular noun and an adjective.



Incorrect Data flow

- Spontaneous generation. The APPLY INSURANCE PREMIUM process, for instance, produces output, but has no input data flow. Because it has no input, the process is called a spontaneous generation process.
- Black hole. The CALCULATE GROSS PAY is called a black hole process, which is a process that has input, but produces no output.
- Gray hole. A gray hole is a process that has at least one input and one output, but the input obviously is insufficient to generate the output shown. For example, a date of birth input is not sufficient to produce a final grade output in the CALCULATE GRADE process.

Incorrect Data flow

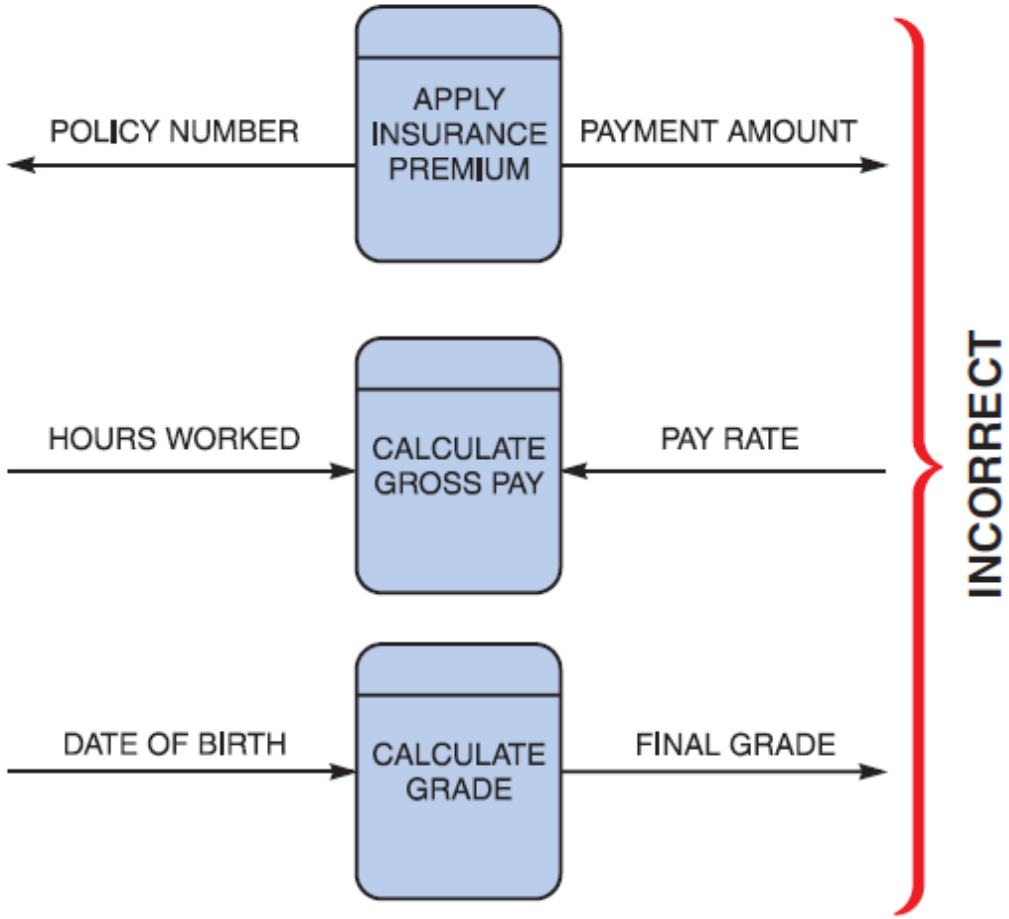


Figure: Caption

DATA STORE SYMBOL

- Similarly, a company stores employee salary and deduction data during the year in order to print W-2 forms with total earnings and deductions at the end of the year.
- A DFD does not show the detailed contents of a data store — the specific structure and data elements are defined in the data dictionary, which is discussed later in this chapter.
- In a DFD, the Gane and Sarson symbol for a data store is a flat rectangle that is open on the right side and closed on the left side. The name of the data store appears between the lines and identifies the data it contains. A data store name is a plural name consisting of a noun and adjectives, if needed. Examples of data store names are STUDENTS, ACCOUNTS RECEIVABLE, PRODUCTS, DAILY PAYMENTS, PURCHASE ORDERS, OUTSTANDING CHECKS, INSURANCE POLICIES, and EMPLOYEES

correct data store

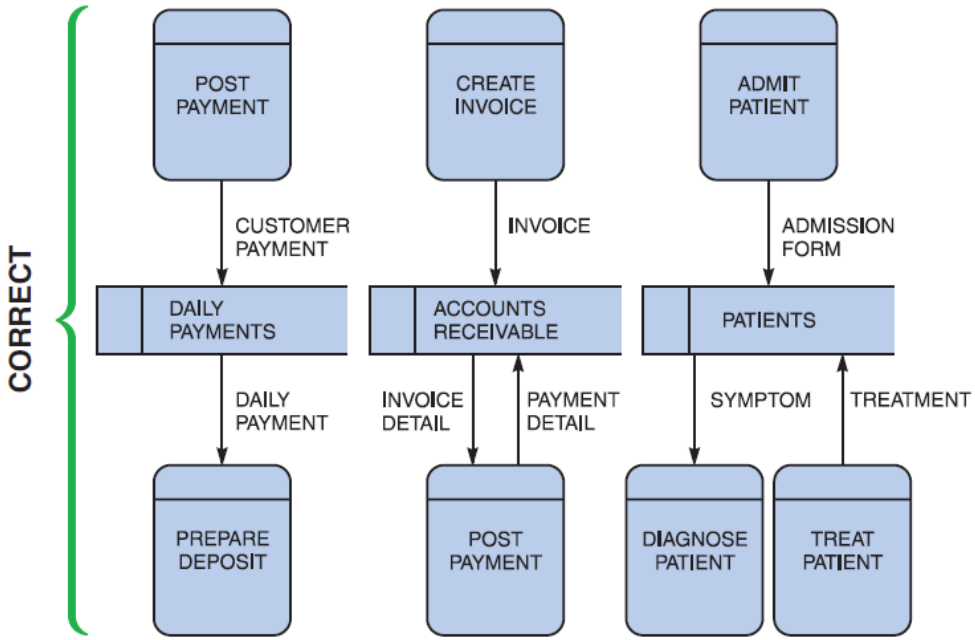
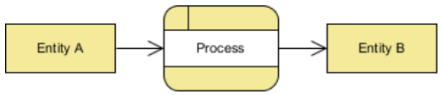
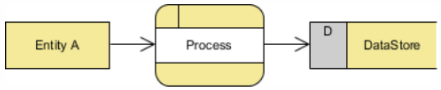
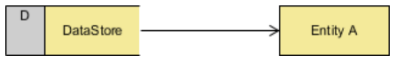
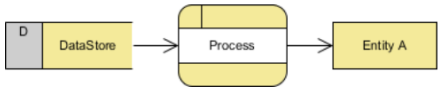
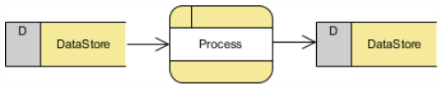


FIGURE 5-7 Examples of correct uses of data store symbols in a data flow diagram.

Figure: Caption

One of the rule for developing DFD is that all flow must begin with and end at a processing step. This is quite logical, because data can't transform on its own with being process. By using the thumb rule, it is quite easily to identify the illegal data flows and correct them in a DFD.

Wrong	Right	Description
		An entity cannot provide data to another entity without some processing occurred.
		Data cannot move directly from an entity to a data story without being processed.
		Data cannot move directly from a data store without being processed.
		Data cannot move directly from one data store to another without being processed.

ENTITY SYMBOL

- The symbol for an entity is a rectangle, which may be shaded to make it look three-dimensional. The name of the entity appears inside the symbol.
- The symbol for an entity is a rectangle, which may be shaded to make it look three-dimensional. The name of the entity appears inside the symbol.
- A DFD shows only external entities that provide data to the system or receive output from the system. A DFD shows the boundaries of the system and how the system interfaces with the outside world. For example, a customer entity submits an order to an order processing system.
- Other examples of entities include a patient who supplies data to a medical records system, a homeowner who receives a bill from a city property tax system, or an accounts payable system that receives data from the company's purchasing system.

Correct Entity

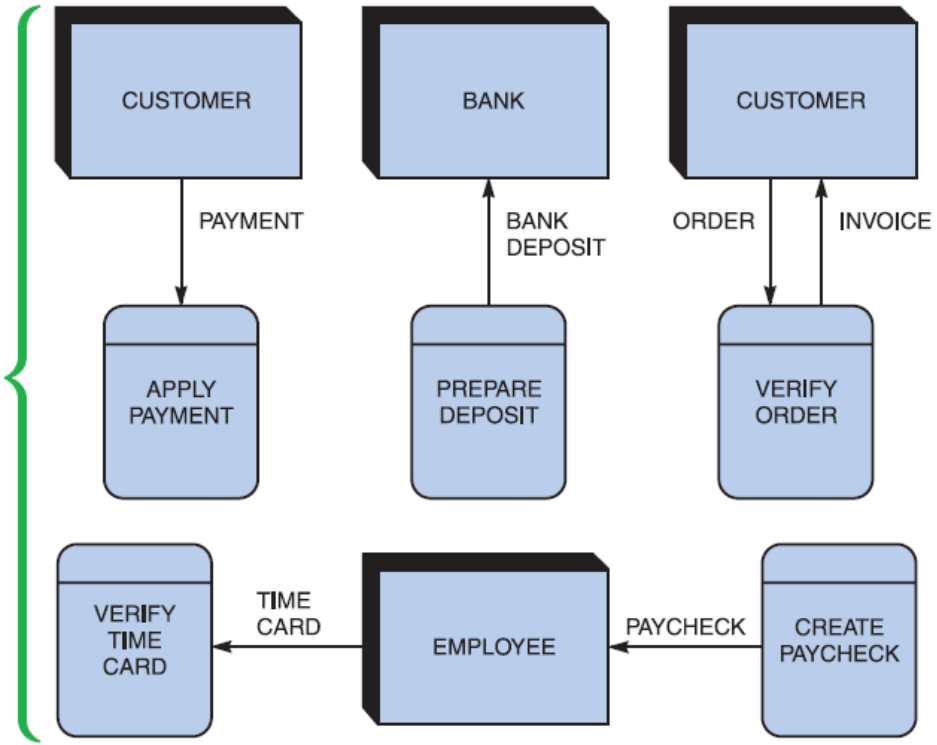


Figure: Examples of correct uses of external entities in a data flow diagram.

incorrect entity

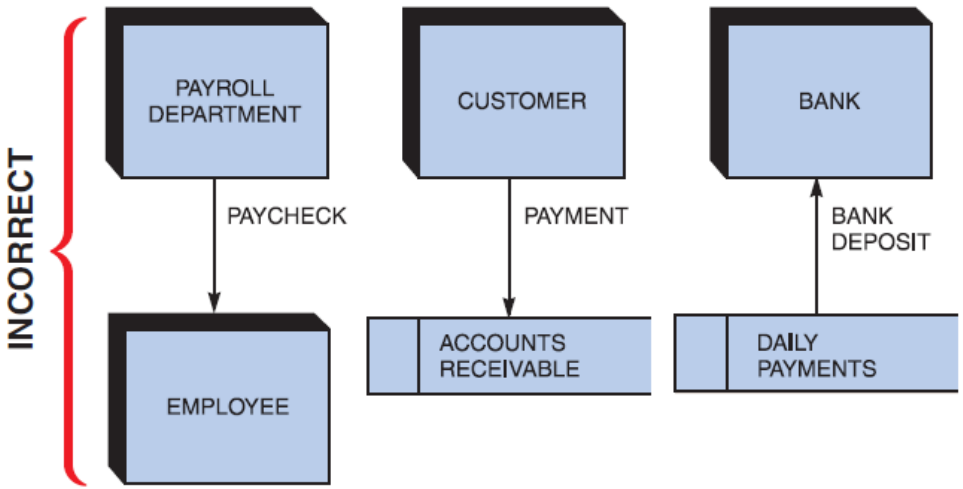


Figure: Examples of incorrect uses of external entities. An external entity must be connected by a data flow to a process, and not directly to a data store or to another external entity.

Correct and Incorrect Examples of Data Flows

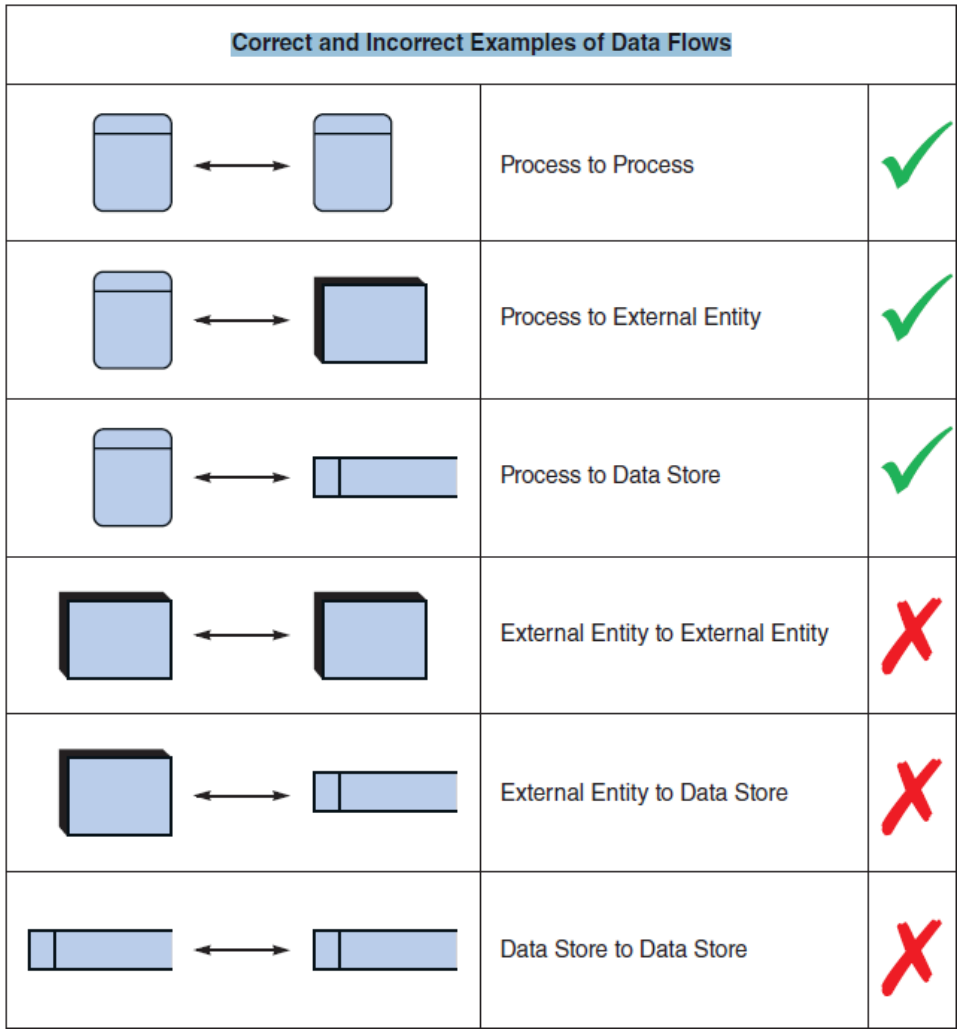


Figure: Examples of correct and incorrect uses of data flows..

Guidelines for Drawing DFDs

When you draw a context diagram and other DFDs, you should follow several guidelines

- Draw the context diagram so it fits on one page,
- Use the name of the information system as the process name in the context diagram. For example, the process name in Figure 5-12 is GRADING SYSTEM. Notice that the process name is the same as the system name. This is because the context diagram shows the entire information system as if it were a single process. For processes in lower-level DFDs, you would use a verb followed by a descriptive noun, such as ESTABLISH GRADEBOOK, ASSIGN FINAL GRADE, or PRODUCE GRADE REPORT.
- Use unique names within each set of symbols. For instance, the diagram in Figure 5-12 shows only one entity named STUDENT and only one data flow named FINAL GRADE. Whenever you see the entity STUDENT on any other DFD in the grading system, you know that you are dealing with the same entity. Whenever the FINAL GRADE data flow appears, you know that you are dealing with the same data flow. The naming convention also applies to data stores.
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continue

- Do not cross lines. One way to achieve that goal is to restrict the number of symbols in any DFD. On lower-level diagrams with multiple processes, you should not have more than nine process symbols. Including more than nine symbols usually is a signal that your diagram is too complex and that you should reconsider your analysis. Another way to avoid crossing lines is to duplicate an entity or data store. When duplicating a symbol on a diagram, make sure to document the duplication to avoid possible confusion. A special notation, such as an asterisk, next to the symbol name and inside the duplicated symbols signifies that they are duplicated on the diagram.
- Provide a unique name and reference number for each process. Because it is the highest-level DFD, the context diagram contains process 0, which represents the entire information system, but does not show the internal workings. To describe the next level of detail inside process 0, you must create a DFD named diagram 0, which will reveal additional processes that must be named and numbered. As you continue to create lower-level DFDs, you assign unique names and reference numbers to all processes, until you complete the logical model.
- Obtain as much user input and feedback as possible. Your main objective is

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Any question??

